

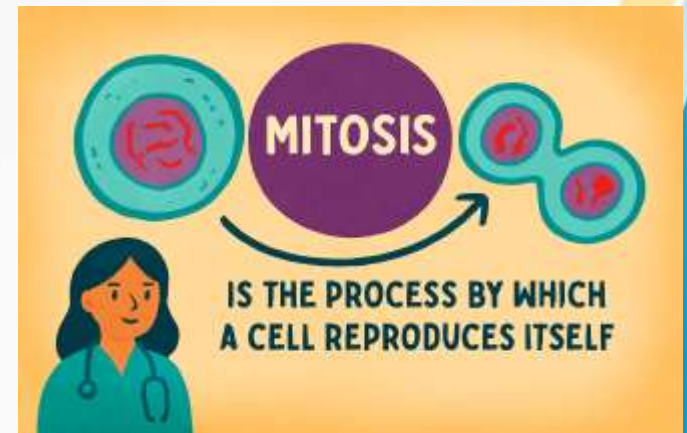
CARE OF ONCOLOGY PATIENTS



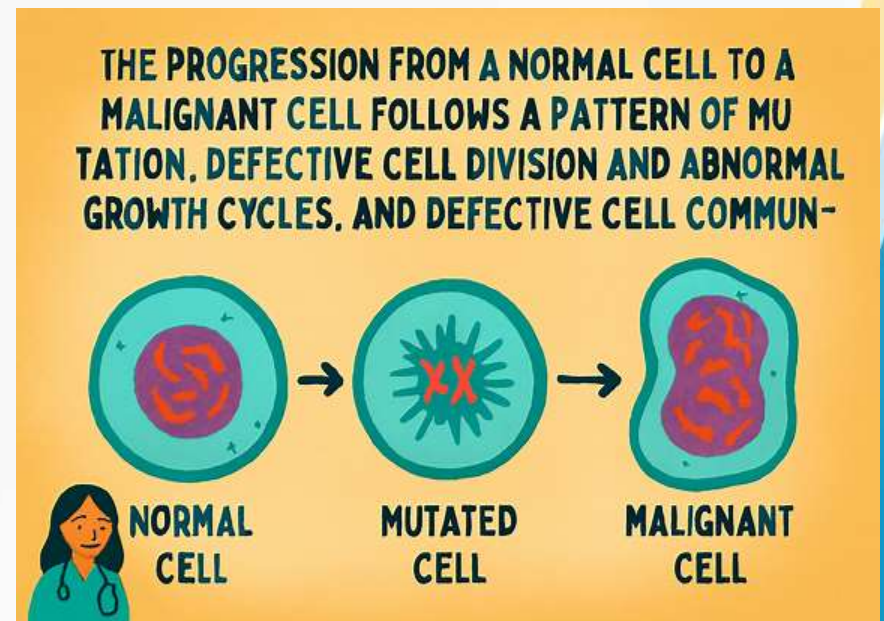
Dr/ Zinab Mohamed

Physiology of Cancer

- **Mitosis** is the process by which a cell reproduces itself. Mitosis is necessary for the growth to replace those that are worn out and to repair damage done to cells by illness and injury.



- The progression from a normal cell to a malignant cell follows a pattern of mutation, defective division and abnormal growth cycles, and defective cell communication.



ONCOLOGY NURSE



- **Oncology:** is the branch of medicine dealing with tumors.
- **Oncology nurses** specialize in the care, treatment of clients with cancer.

☞ **Neoplasm** is a term that combines the Greek word neo, meaning —new,|| and plasia, meaning —growth,|| to suggest new tissue growth. (new growth of tissue, or tumor), which is not beneficial

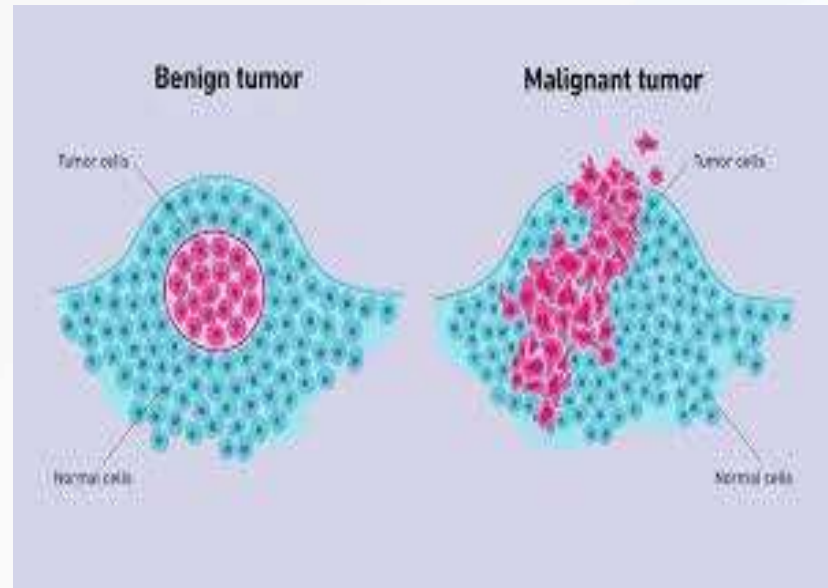
☞

- ☞ The new growth results in enlargement of tissue and the formation of an abnormal mass.

Benign indicates a neoplasm that is usually harmless.

Benign growths are almost always **encapsulated**
(surrounded by a capsule).

- **Malignant** (uncontrolled growth that can lead to death) growths are quite different from normal cells. Cells that exhibit malignant growth are known as cancer.



Classification of Tumor

Tumor can be classified according to behavior into benign, malignant.

Benign

- Slow, steady growth.
Localized and capsulated.
- Smooth, well-defined and
Movable when palpated.
- Recurs rare.
- Resembles body tissue.

Malignant

- Growth rapidly.
- Metastasis
- Irregular and
immobile.
- May be recurs.
- Little
resemblance.

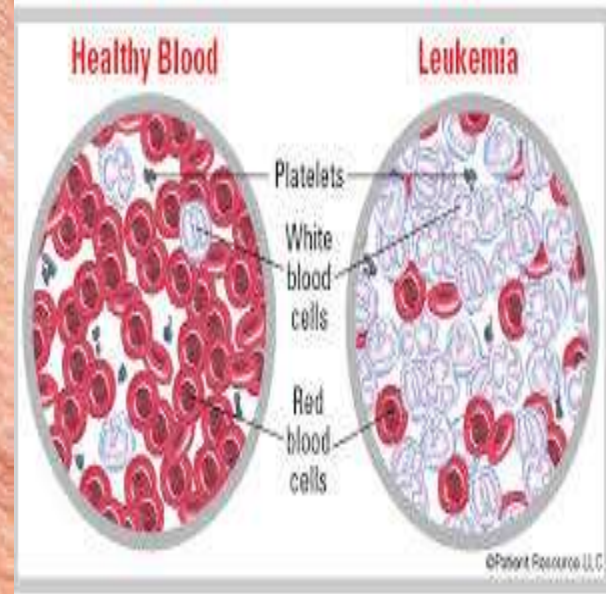
Classification of Tumor Cells

- Cancers are **identified by** the tissue affected, speed of cell growth, cell appearance, and location.
- Tumors are often classified **according to** the organs or tissues from which they first began to grow

- The suffix **-oma** means —tumor; this suffix is used in the names of various kinds of growths or swellings either fluid (**hematoma**) or swelling containing malignant cells(**lymphoma**)

- A **carcinoma** originates in epithelial tissues (skin and mucous membranes, stomach, uterus, lungs, skin, and tongue).
- A **sarcoma** arises from mesenchymal tissues (bone, and muscles).

- **Leukemia** and **lymphoma** are cancers of the blood-forming system.
- **Melanoma** malignancy of the pigment cells of the skin



Metastasis refers to the movement of malignant cells from the site of origin to another site.

Cancer cells are able to: invade blood or lymph vessels and grow in a new location.

- **Primary tumor:** the original site of the malignancy
- **Secondary tumor:** areas where malignancy has spread i.e. metastasis Common sites of metastasis are lymph nodes, liver, lungs, bones, brain

Causative Factors

- All cancer results from defects in the DNA of genes; either **inherited** or by **mutation** (a permanent change in the DNA sequence of a gene).
- Each person's body has a different ability to resist the effects of cancer- causing substances (**carcinogens**).

Chemical Carcinogens

- Some pesticides
- Chewing tobacco
- Cigarette smoking
- Immunosuppressive drugs



Physical Carcinogens

- ✓ **Radiation** may originate from x-ray machines and radioactive elements or from the ultraviolet rays of the sun can produce **skin cancer**.
- ✓ **Viruses:** The hepatitis B&C virus is carcinogenic for **liver cancer**

- Only 5% to 10% of cancers are related to a directly inherited gene.

Contributing Factors

- **Intrinsic Factors:** Age, and sex
- **Stress:** Stress over long periods
- **Obesity:** Cancers of the breast, colon, rectum, liver, and kidney are all linked to excess weight and obesity.

Diet

- People who eat high-fat.
- Consumption of large amounts of smoked foods.
- A diet low in vitamins A, C, and E.

Hormones

- Long-term use of the female hormone estrogen

Measures to Prevent Cancer

Each nurse must teach patients the following measures at every opportunity:

- Eat a varied diet and balance caloric intake with exercise
- Limit red meat and processed meat intake.
- Eat fatty fish twice a week to increase omega-3 intake.
- Avoid smoked, and salt-cured foods.

- 
- ☞ Learn and regularly practice self-examination techniques.
 - ✓ Withhold from smoking or using tobacco products.
 - ✓ Avoid overexposure to the sun
 - ✓ Maintain weight within suggested limits.

Effects of Cancer

- a. Anoxia and necrosis of organs
- b. Loss of function: bowel or bladder obstruction
- c. Lung cancer: respiratory difficulties
- d. Airway obstruction
- e. Disturbed liver functioning (Ascites)
- f. Motor and sensory deficits (compresses nerves)

- 
- Abnormal WBCs: impaired immunity
 - Diminished RBCs and platelets: anemia and clotting disorders
 - Hemorrhage and severe anemia
 - Unexplained rapid weight loss

SEVEN WARNING SIGNALS OF CANCER



**Change in
bowel habits
or bladder
function**



**Sores that
do not heal**



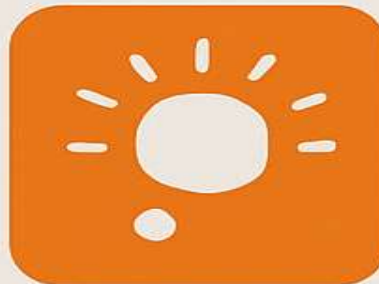
**Unusual
bleeding or
discharge**



**Thickening or
lump in the breast
or other body
parts**



**Indigestion or
difficulty
swallowing
persistently**



**Obvious
change in a
wart or mole**



**Nagging cough
or hoarseness
of voice**

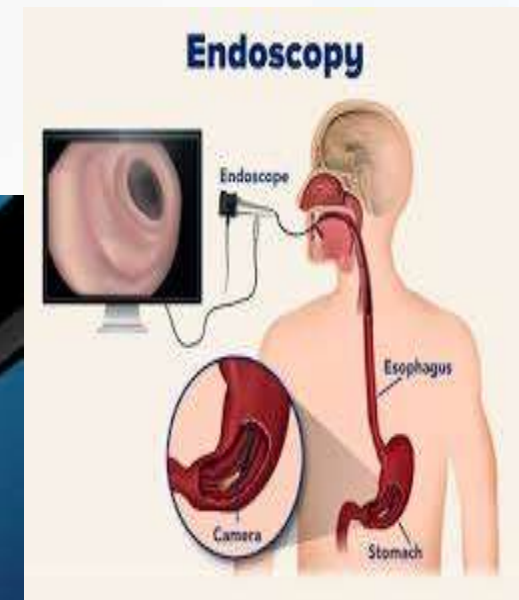
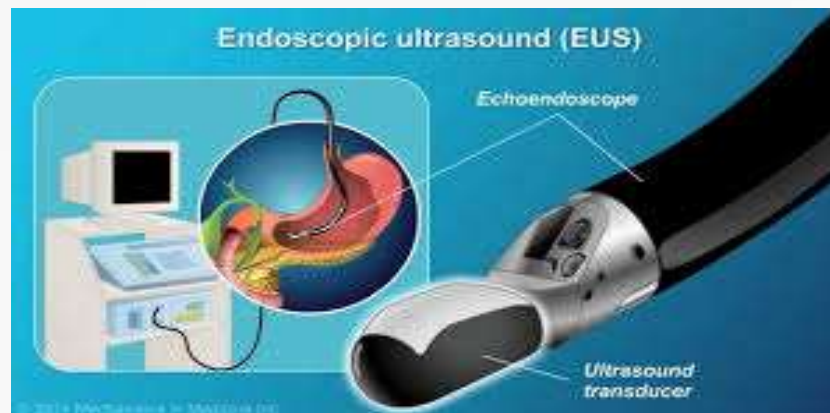
Diagnostic test:

Non Invasive Diagnostic

- ✓ X-ray film
- ✓ Magnetic resonance imaging: MRI & CT scan
- ✓ Mammography
- ✓ Barium enema
- ✓ Stool tests
- ✓ A complete blood count

Invasive Diagnostic Techniques

- Endoscopy
- Endoscopy with ultrasound:
- Biopsy
- Tumor markers



TNM staging system identifies cancers by how much the malignancy has spread

- ☞ The three basic parts of the system are *T* for primary tumor, *N* for regional nodes, and *M* for metastasis.

Classification	Staging	Tissue Involvement
Primary tumor (T) T_{is}	Stage I	Tumor in situ; indicates no invasion of other tissues
T_1, T_2, T_3, T_4	Stage II	Ranges indicate progressive increase in tumor size with local metastasis
Regional lymph node involvement (N) N_0 $N_1, N_2, N_3,$	No nodes Stage III	Metastasis to regional lymph nodes
Metastasis (M) M_0 M_1	No metastasis Stage IV	Distant metastasis

Common Nursing Problems for Patients with Cancer

- . Altered nutrition status.
- . Potential for infection.
- . Pain, acute or chronic.
- . Altered skin integrity.
- . Altered body image.

- Altered physical mobility.
- Diarrhea and dehydration.
- Constipation
- Altered urinary elimination
- Fatigue

- 
- . Insufficient knowledge of drugs and their potential side effects
 - . Altered self-care ability.
 - . Fear of dying.

Multiple modalities are commonly used in cancer treatment

- ♦ Surgery
- ♦ Radiation therapy
- ♦ Chemotherapy
- ♦ Hormone therapy
- ♦ Hyperthermia
- ♦ Acupuncture
- ♦ Gene therapy
- ♦ Bone marrow transplantation and stem cell

Surgery

Surgical removal of the entire cancer remains the ideal treatment method.

Types of cancer surgeries

- Diagnostic Surgery
- Curative \ Treatment
- Prophylactic Surgery
- Palliative Surgery

- **Diagnostic Surgery:** such as a **biopsy**
- **Surgery as curative:**

Remove the entire tumor or as much as is feasible and may involve surrounding tissue, OR regional lymph nodes.

- **Prophylactic Surgery:** involves removing tissues or organs that are likely to develop cancer.
- **Palliative Surgery:** an attempt to relieve complications of cancer, such as obstructions.

Radiation Therapy

- The purpose of radiation is to destroy malignant cells without permanent damage to adjacent body tissues.

Different Forms of Radiation

1. **Electromagnetic radiations:** ultraviolet and and infra- red rays.
2. **Particulate radiations:** produced by alpha and beta particles.
3. **Atomic radiation** and nuclear exposures.

Effect of radiation

- **A- Local effect**

- ◆ **Radiation sickness** causing nausea, vomiting, and depression
- ◆ **Radiation cachexia** caused by poor appetite, weight loss, and anemia
- ◆ **Anemia**, and leucopenia

Systemic effect

Skin

- ❑ Erythema , Edema , Chronic ulcer
- ❑ Loss of hair

Bone and cartilage

- ◆ Bone necrosis
- ◆ pathological fracture

GIT

- Increase mucus Ulceration and hemorrhage
- Atrophy of lymphoid tissue, and Necrosis

Kidney

- Radiation nephritis
- Chronic renal failure

Testes and ovaries

- Sterility

Bone marrow and blood

- Anemia
- Leucopenia
- Thrombocytopenia
- Leukemia

Nursing Management in Radiation Therapy

Nursing protection from radiation



Time



Shield

sealed sources of radiation implanted

Do not remain within 1meter of the patient any longer than required to give essential care.

Notify the radiation therapist of any implant that has moved out of position

Do not discard any dressings or linens

**After the patient is discharged
from the hospital,
the radiologist check the room
with survey meter**



Instruct the patient on external radiation therapy to:

AVOID washing the marked skin with SOAP, JUST APPLY WATER ONLY.



RADIATION THERAPY

**AVOID USING DEODORANT, LOTION,
MEDICATION, PERFUMES TO THE SITE
DURING THE TREATMENT PERIOD.**



DEODORANT



LOTION



MEDICATION



PERFUMES

AVOID RUBBING, SCRATCHING OR SCRUB TREATED AREA



RUBBING



SCRATCHING



SCRUB

**AVOID APPLYING HEAT NOR
COLD TO THE TREATED AREA.**



HEAT



COLD

**Protect
skin from
sun
exposure**

**Be sure
to get plenty
of rest**

Common Side Effects of Radiation Therapy based on site

1-Head and Neck

Mucositis



Esophagitis



Nausea & Indigestion

Chest

The diagram consists of three vertical columns. Each column has a blue arrow pointing downwards from a yellow label at the top to a colored rectangular box below. The first column is green, the second is blue, and the third is grey. The background is light grey with a decorative blue and yellow curved shape on the right side.

**Nausea,
vomiting,
diarrhea**

Abdomen

**Inflammation
of lung
tissue**

Pelvis

**.Urethral and
rectal stenosis**

General Side Effects





CHEMOTHERAPY

- **Chemotherapy** is the use of anticancer drugs to treat cancerous cells.
- **Chemotherapy** works by interfering with the cancer cell's ability to grow or reproduce.

Goals of Chemotherapy



Cure



Control



Palliation

Side effects of Chemotherapy

↓ **Red
blood cells**

- ↓ **White
blood
cells**

- ↓ **Platelet
s**

Nausea

- **Vomitin
g**

Diarrhea

- **Alopeci
a**

SAFETY MEASURES IN HANDLING CHEMOTHERAPY



Wear powder-free latex gloves when preparing or working with chemotherapeutic agents.



Wash hands before putting on and after removing gloves.



Change latex gloves after each use, tear, puncture, or medication spill

Wear a long-sleeved, nonabsorbent gown with elastic at the wrists and back closure.



Eye and face shields should be worn

Wrap gauze or alcohol pads around the neck of ampules when opening



Wrap gauze or alcohol pads around injection sites when removing syringes or needles from IV injection ports.



Label all syringes and IV tubing containing chemotherapeutic agents as hazardous material.



Place an absorbent pad directly under the injection site to absorb any accidental spillage.



If any contact with the skin occurs, immediately wash the area thoroughly

- **Hormonal Therapy:** are chemicals produced by glands in the body and slows the progress of the tumor.
- **Gene therapy:** Is a therapeutic technique in which a functioning gene is inserted into a cell to correct a metabolic abnormality or to introduce a new function.

Bone Marrow and Stem Cell Transplantation

➤ Types of Transplants

1. "**Allogeneic**" **BMT**, if bone marrow from a donor is used
2. "**Syngeneic**" **BMT**, if the donor is an identical twin.
3. **Autologous BMT**, patients may be their own bone marrow donors.
 - ✓ **mainly used to correct the severe bone marrow damage**

- **Acupuncture** controls symptoms caused by cancer treatment, such as weight loss, cough, chest pain, fever, anxiety, depression, night sweats, hot flashes, and dry mouth.



BIOLOGIC RESPONSE MODIFIERS (BRMS)

are agents that manipulate the immune system in the hope of controlling or curing malignancy with little or no toxic effect on normal cells.



EXAMPLES OF COMMON PROBLEMS RELATED TO CANCER OR CANCER TREATMENT



ANOREXIA



MUCOSITIS



WEIGHT LOSS



**NAUSEA, VOMITING
AND DIARRRHEA**

Clinical Scenario

- Mr. Youssef, a 65-year-old retired teacher, has been diagnosed with laryngeal cancer. He is undergoing external radiation targeting the head and neck region. After two weeks of treatment, he reports oral pain, difficulty swallowing, and loss of taste. His skin over the treated area appears dry and slightly reddened. He feels fatigued and has lost 3 kg since starting therapy.

Which nursing intervention is most appropriate for Mr. Youssef's skin care during external radiation?

- A. Apply alcohol-based lotion to the treated area
- B. Use deodorant to reduce odor
- C. Avoid scrubbing and use only water on marked skin
- D. Apply cold compresses daily

- **Correct Answer: C**

Which dietary advice is most helpful for Mr. Youssef?

- A. Eat spicy foods to stimulate taste
- B. Avoid all fluids during meals
- C. Choose soft, high-protein, high-calorie foods
- D. Fast before radiation sessions

- **Correct Answer: C**

Thank
You